You

I have the following network in Cisco packet tracer (attached screenshot [Expired]), how can I do this, what could be wrong (explain all the possibilities as if you are a computer science professor who specialises in computer networks)?

Download the file named "Network3-TS.pkt" file.
All PCs should be able to communicate with each other without any issues.
However, the network is experiencing the following issues.
Your task is to troubleshoot and resolve these issues.

- *Issue 1: PC2 and PC3 are unable to communicate with each other.*
- *Issue 2: Neither PC2 nor PC3 can communicate with PC0 and PC1.*
- *Issue 3: Neither PC0 nor PC1 can communicate with PC4 or PC5.*

Copilot

It seems like a classic network troubleshooting scenario—let's approach this systematically, step by step, and see where the issues might lie. Here's an in-depth analysis of potential causes and solutions for each issue:

Issue 1: PC2 and PC3 are unable to communicate with each other.
Possible Causes:

1. **IP Configuration Problems:**
   - PC2 and PC3 might not be in the same subnet. For instance, their IP addresses or subnet masks might not align.
   - The default gateways might be misconfigured or missing.
2. **Switch Configuration Issues:**
   - The ports connected to PC2 and PC3 might not belong to the same VLAN, causing the communication to be blocked.
   - The switchport settings (e.g., duplex or speed mismatches) could be incorrectly set.
3. **Physical Layer Problems:**
   - Cables connecting PC2 and PC3 to the switch might be faulty or disconnected.

Steps to Resolve:

1. Check IP addresses, subnet masks, and default gateway configurations on both PCs to ensure they match the same network.
2. Log into the switch and verify VLAN assignments on the connected ports using commands like show vlan brief.
3. Test connectivity using the ping command between PC2 and PC3.
4. Check the physical connections and the switchport status using show interface status.

Issue 2: Neither PC2 nor PC3 can communicate with PC0 and PC1.
Possible Causes:

1. **Routing Table Misconfigurations:**
   - The routers might not have routes configured to pass packets between the subnet of PC2/PC3 and the subnet of PC0/PC1.
2. **Access Control Lists (ACLs):**
   - ACLs on the routers could be blocking communication between the subnets.
3. **Serial Connection Problems:**
   - The serial links between the routers might be down due to misconfigured or mismatched IP settings, encapsulation protocols, or clock rates.
4. **NAT/Firewall Issues:**
   - If Network Address Translation (NAT) or a firewall is in use, it might be improperly configured, preventing communication between the subnets.

Steps to Resolve:

1. Verify routing configurations on each router using show ip route. Add static routes if required or configure dynamic routing protocols (e.g., RIP, OSPF, or EIGRP).
2. Inspect ACLs on the routers using show access-lists and ensure they are allowing traffic between the subnets.

3. Check the status of the serial interfaces using show interface serial 0/0/0 and show ip interface brief. Confirm that the IP addresses and encapsulation protocols match.
4. Test connectivity across routers using ping from one router's interface to the other.

Issue 3: Neither PC0 nor PC1 can communicate with PC4 or PC5.
Possible Causes:

1. **Routing or IP Configuration Errors:**
   - Similar to Issue 2, routing tables might lack the necessary routes to connect the subnets.
2. **Switch Problems:**
   - Switches connecting PC0/PC1 and PC4/PC5 might have VLAN mismatches or trunking misconfigurations.
3. **Serial or Layer 2 Problems:**
   - The serial connections or intermediary devices (e.g., switches) might have misconfigurations or be down.

Steps to Resolve:

1. Check and configure routing on the routers to ensure they know how to reach the respective subnets for PC0/PC1 and PC4/PC5.
2. Examine switch configurations for PC4 and PC5 to ensure the VLANs and trunks are properly set up. Use commands like show vlan and show interfaces trunk.
3. Test connectivity across the network's serial connections, ensuring no layer 2 or physical issues.

General Recommendations:

- Use ping and traceroute commands extensively to isolate where communication breaks down.
- Examine device logs (show logging) for error messages on routers or switches.
- If a dynamic routing protocol is in use, confirm it is running correctly on all routers and that the networks have been advertised.
- Ensure all interfaces are up and configured with the correct IP addresses and subnet masks.
- Check for any overlapping IP address ranges across different subnets that could cause issues.

Would you like me to elaborate on any specific aspect or help with configurations? Let me know—I'm here to dig deeper!

You

I'm trying to resolve issue 2, this is my output:

%SYS-5-CONFIG_I: Configured from console by console
show interface serial 0/0/0
Serial0/0/0 is up, line protocol is up (connected)
Hardware is HD64570
Internet address is 12.5.6.3/8
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
Conversations 0/0/256 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up

Router#
Router#show ip interface brief

Interface	*IP-Address*	*OK? Method Status*	*Protocol*
FastEthernet0/0	*192.167.55.2*	*YES manual up*	*up*
FastEthernet0/1	*unassigned*	*YES unset administratively down*	*down*
Serial0/0/0	*12.5.6.3*	*YES manual up*	*up*
Serial0/0/1	*14.5.7.3*	*YES manual up*	*up*
Vlan1	*unassigned*	*YES unset administratively down*	*down*

Copilot

Great work in gathering these outputs—they're crucial for diagnosing Issue 2. Let's analyze this step by step based on what you've shared:

Observations

1. **Serial Interface (Serial0/0/0):**
   - The interface is **up**, and the line protocol is **up (connected)**, which is a good sign. This indicates that the physical and data link layers are operational.
   - **Encapsulation:** It's set to HDLC. If this matches the encapsulation on the other end of the serial connection, then this shouldn't be an issue.
   - **IP Address:** The configured IP (12.5.6.3) with a /8 mask seems fine. However, we need to verify whether the other router's serial interface IP is in the **same subnet** (e.g., 12.0.0.0/8).
2. **FastEthernet0/0:**
   - This interface is also **up**, with an IP address of 192.167.55.2. Ensure that this IP and subnet mask match the devices connected to this interface.
3. **FastEthernet0/1:**
   - It's administratively **down**, so it won't be used for routing. If you're not using it, this is fine.
4. **VLAN1 Interface:**
   - This is **administratively down**, which is expected unless specifically configured as a layer-3 interface.

Potential Causes and Resolutions

1. Subnet Mismatch Across Serial Connections

- Ensure that the router's **Serial0/0/0** on the other end of this link has an IP address in the same subnet as 12.5.6.3/8. Use the show ip interface brief or show running-config commands on the other router to verify this.
- If there's a mismatch, reconfigure the IP address on the other end to match the 12.0.0.0/8 network. Example:
- config terminal
- interface serial 0/0/0
- ip address 12.5.6.4 255.0.0.0
- no shutdown

2. Missing Static Route or Dynamic Routing Protocol

- If you're using **static routing**, verify that a route exists for traffic between the networks connected to this router and the networks connected to the other router. Use the show ip route command.
  - For example, if PC0 and PC1 are on 192.167.55.0/24 and the other router's network is on a different subnet (e.g., 192.168.1.0/24), you need a static route:
  - ip route 192.168.1.0 255.255.255.0 12.5.6.4
  - Add a corresponding static route on the other router pointing back to 192.167.55.0/24.
- Alternatively, enable a **dynamic routing protocol** (e.g., RIP, OSPF, or EIGRP) to automatically exchange routes between the routers:
  - For RIP:
  - config terminal
  - router rip
  - version 2
  - network 12.0.0.0
  - network 192.167.55.0
  - no auto-summary

3. Access Control Lists (ACLs) Blocking Traffic

- Check for any ACLs applied to the serial interfaces or on the routers themselves that might block traffic. Use show access-lists or show running-config to confirm.
- If there's an ACL blocking traffic, either modify it or temporarily remove it for testing.

4. Issues with the Other Serial Interface (Serial0/0/1)

- Your Serial0/0/1 is up with an IP of 14.5.7.3/8, but it might be leading to a separate network. Make sure there's no confusion between Serial0/0/0 and Serial0/0/1 when configuring routes.

Next Steps for Testing

1. **Test Layer 3 Connectivity:**
   - Use the ping command to test from this router's Serial0/0/0 interface to the other router's serial interface IP.
2. **Verify Routes:**
   - Run show ip route on both routers and ensure there are routes for all relevant networks.
3. **Traceroute:**
   - Use traceroute from one PC to another to identify where packets are getting dropped.
4. **Loopback Testing:**
   - If needed, temporarily set up a loopback interface on one router and configure a route to it to test isolated routing behavior.

Would you like detailed commands for any of these steps or guidance on enabling specific routing protocols? Let me know!

You

This is the output from router 0:

Router#
%SYS-5-CONFIG_I: Configured from console by console
show interface serial 0/0/0
Serial0/0/0 is up, line protocol is up (connected)
Hardware is HD64570
Internet address is 12.5.6.2/8
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
Conversations 0/0/256 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up

Router#show ip interface brief

Interface	*IP-Address*	*OK? Method Status*	*Protocol*
FastEthernet0/0	*192.167.54.1*	*YES manual up*	*up*
FastEthernet0/1	*unassigned*	*YES unset administratively down*	*down*
Serial0/0/0	*12.5.6.2*	*YES manual up*	*up*
Serial0/0/1	*unassigned*	*YES unset administratively down*	*down*
Vlan1	*unassigned*	*YES unset administratively down*	*down*

router 0 is connected to pc0 and pc1 through a switch on port fa0/0 and default gateway 192.167.54.1.
router 1 is connected to pc2 and pc3 through a switch on port fa0/0 and default gateway 192.167.55.2

Copilot

This is helpful information—thank you. Based on the configuration and output provided for **Router 0**, let's examine potential causes and solutions to resolve **Issue 2** (PC2 and PC3 not communicating with PC0 and PC1).

Analysis

1. **Serial0/0/0 Interfaces on Router 0 and Router 1:**
   - Router 0's Serial0/0/0 has the IP address 12.5.6.2/8.
   - Router 1's Serial0/0/0 (from the previous output) has the IP address 12.5.6.3/8.
   - Both IPs are within the same subnet 12.0.0.0/8, and both interfaces are up. This indicates that **Layer 3 connectivity between the routers is established**.
2. **FastEthernet0/0 Interfaces:**
   - Router 0 is connected to the subnet 192.167.54.0/24 (PC0 and PC1).
   - Router 1 is connected to the subnet 192.167.55.0/24 (PC2 and PC3).
   - The default gateways for PCs seem to be configured correctly (192.167.54.1 for PCs on Router 0 and 192.167.55.2 for PCs on Router 1).
3. **Routing Issues:**
   - The most likely issue here is that **Router 0 and Router 1 don't know about each other's LAN subnets** (192.167.54.0/24 and 192.167.55.0/24), causing packets to drop after traversing the serial link.

Steps to Resolve

1. Verify Routing Table

- On both routers, verify the routing table using the show ip route command. Ensure that:
  - Router 0 knows about 192.167.55.0/24 (connected to Router 1).
  - Router 1 knows about 192.167.54.0/24 (connected to Router 0).
- If these routes are missing, packets sent from PC0/PC1 to PC2/PC3 (or vice versa) won't reach their destination.

2. Add Static Routes

- If dynamic routing protocols are not enabled, add **static routes** on both routers to manually define how to reach the other LAN.

On Router 0:

config terminal
ip route 192.167.55.0 255.255.255.0 12.5.6.3
end

On Router 1:

config terminal
ip route 192.167.54.0 255.255.255.0 12.5.6.2
end

3. Use a Dynamic Routing Protocol

- Alternatively, enable a dynamic routing protocol (like RIP, OSPF, or EIGRP) on both routers. This allows them to automatically exchange routes.

Example using RIP on both routers:

config terminal
router rip
version 2
network 12.0.0.0
network 192.167.54.0 ! On Router 0
network 192.167.55.0 ! On Router 1
no auto-summary
end

- Verify the RIP configuration using show ip protocols and check if the routers are exchanging routes.

4. Test Connectivity

- Use the ping command on Router 0 to test connectivity to 192.167.55.2 (Router 1's FastEthernet0/0).
- Similarly, use ping on Router 1 to test connectivity to 192.167.54.1 (Router 0's FastEthernet0/0).
- Perform a traceroute from PC0 to PC2 or PC3 to verify the packet path.

5. ACL/Firewall Check

- Ensure there are no ACLs or firewalls applied on the routers that are blocking traffic between the subnets. Use show access-lists or inspect the running configuration.

Expected Result After Fix

- With the routing issues resolved, PC0 and PC1 should be able to communicate with PC2 and PC3. If there are any remaining issues, focus on testing the switches and PCs for VLAN or IP misconfigurations.

Let me know how the testing goes, and if further troubleshooting is needed!

You

(Now Thursday, Mar 27)
How about fixing issue 3 which is linking PC0 and 1 to PC4 and 5. PC0 and 1 are connected to router 0 and PC4 and 5 are connected to router 2. Output from router 0:

Router#
%SYS-5-CONFIG_I: Configured from console by console
show interface serial 0/0/0
Serial0/0/0 is up, line protocol is up (connected)
Hardware is HD64570
Internet address is 12.5.6.2/8
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
Conversations 0/0/256 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up

Router#show interface serial 0/0/1
Serial0/0/1 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
Conversations 0/0/256 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 2 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.167.54.1    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            12.5.6.2        YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Output of router 2:

```
Router#
%SYS-5-CONFIG_I: Configured from console by console
show interface serial 0/0/0
Serial0/0/0 is up, line protocol is up (connected)
 Hardware is HD64570
 Internet address is 14.5.7.2/8
 MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
 Encapsulation HDLC, loopback not set, keepalive set (10 sec)
 Last input never, output never, output hang never
 Last clearing of "show interface" counters never
 Input queue: 0/75/0 (size/max/drops); Total output drops: 0
 Queueing strategy: weighted fair
 Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations  0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
 5 minute input rate 0 bits/sec, 0 packets/sec
 5 minute output rate 5 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    4 packets output, 204 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
    DCD=up  DSR=up  DTR=up  RTS=up  CTS=up

Router#show interface serial 0/0/1
Serial0/0/1 is administratively down, line protocol is down (disabled)
 Hardware is HD64570
 MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
 Encapsulation HDLC, loopback not set, keepalive set (10 sec)
 Last input never, output never, output hang never
 Last clearing of "show interface" counters never
 Input queue: 0/75/0 (size/max/drops); Total output drops: 0
 Queueing strategy: weighted fair
 Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations  0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
 5 minute input rate 0 bits/sec, 0 packets/sec
 5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
    DCD=down  DSR=down  DTR=down  RTS=down  CTS=down
```

Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.167.56.3 YES manual up up
FastEthernet0/1 unassigned YES unset administratively down down
Serial0/0/0 14.5.7.2 YES manual up up
Serial0/0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down

Copilot

To resolve **Issue 3**, where PC0 and PC1 (connected to Router 0) are unable to communicate with PC4 and PC5 (connected to Router 2), let's break it down systematically based on the outputs provided.

Analysis
Serial Connections:

1. **Router 0 Serial0/0/0**:
   - IP: *12.5.6.2/8* and status: **up/up (connected)**.
   - This interface seems to connect Router 0 to Router 1, so it's not directly involved in linking Router 0 to Router 2.
2. **Router 0 Serial0/0/1**:
   - IP: **unassigned**, status: **administratively down**.
   - This could potentially be the interface connecting Router 0 to Router 2. However, it's currently shut down and hasn't been configured with an IP address.
3. **Router 2 Serial0/0/0**:
   - IP: *14.5.7.2/8* and status: **up/up (connected)**.
   - This interface seems operational and is likely involved in connecting Router 2 to Router 1 or Router 0.
4. **Router 2 Serial0/0/1**:
   - IP: **unassigned**, status: **administratively down**.
   - This interface might also be unused or need configuration.

Current Problem:

- **Serial connectivity between Router 0 and Router 2 is missing.**
- **Routing between Router 0's LAN (*192.167.54.0/24*) and Router 2's LAN (*192.167.56.0/24*) has not been established.**

Steps to Resolve
1. Enable and Configure Router 0's Serial0/0/1
Since Serial0/0/1 is administratively down, let's bring it up and assign an IP address:
On Router 0:
config terminal
interface serial 0/0/1
ip address 14.5.7.1 255.0.0.0
no shutdown
end
This assigns it an IP address within the same subnet as Router 2's Serial0/0/0 (*14.0.0.0/8*).
2. Verify Connectivity
After enabling Serial0/0/1 on Router 0, test connectivity to Router 2's Serial0/0/0 using the *ping* command:
ping 14.5.7.2
Ensure that both ends are operational (up/up) and can communicate.
3. Add Static Routes
Next, configure static routes on both routers to enable communication between their LANs:
On Router 0 (to reach PC4 and PC5 via Router 2):
config terminal
ip route 192.167.56.0 255.255.255.0 14.5.7.2
end
On Router 2 (to reach PC0 and PC1 via Router 0):
config terminal
ip route 192.167.54.0 255.255.255.0 14.5.7.1
end
These routes ensure that packets destined for the other LAN are forwarded through the serial connection.

4. Test Routing
Verify routing tables on both routers using *show ip route*. Ensure that:

- Router 0 has a route to *192.167.56.0/24*.
- Router 2 has a route to *192.167.54.0/24*.

Test connectivity between the PCs using the ping command.
5. Enable Dynamic Routing Protocol (Optional)
If dynamic routing is preferred, configure RIP, OSPF, or EIGRP on both routers. For example, with RIP:
On Router 0:
config terminal
router rip
version 2
network 14.0.0.0
network 192.167.54.0
no auto-summary
end
On Router 2:
config terminal
router rip
version 2
network 14.0.0.0
network 192.167.56.0
no auto-summary
end
Check RIP status using show ip protocols and verify route exchange with show ip route.
6. Final Testing
Perform end-to-end testing:

- From PC0 to PC4 (*ping 192.167.56.x*).
- From PC4 to PC1 (*ping 192.167.54.x*).

If successful, the routing issue is resolved.

Expected Result

Once the serial connection between Router 0 and Router 2 is active and routes are configured, PC0/PC1 should be able to communicate with PC4/PC5 seamlessly.

Let me know if you need help with configurations or testing!

You

Ok these are the outputs of the routers, has it worked?

Router 0:

show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 12.0.0.0/8 is directly connected, Serial0/0/0
C 192.167.54.0/24 is directly connected, FastEthernet0/0
S 192.167.55.0/24 [1/0] via 12.5.6.3

Router 2:

show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 12.0.0.0/8 is directly connected, Serial0/0/0
C 192.167.54.0/24 is directly connected, FastEthernet0/0
S 192.167.55.0/24 [1/0] via 12.5.6.3

Copilot

The outputs you've shared are very insightful, but they indicate that the routing configurations necessary to resolve **Issue 3**—communication between PC0/PC1 (Router 0) and PC4/PC5 (Router 2)—are still incomplete. Let me break it down for you.

Observations
Router 0:

- The routing table only shows:
  - **Connected networks:**
    - *192.167.54.0/24* (LAN connected to FastEthernet0/0 for PC0/PC1).
    - *12.0.0.0/8* (connected via Serial0/0/0 to Router 1).
  - **Static route:** *192.167.55.0/24* via *12.5.6.3* (Router 1).

However, Router 0 is missing a route to **192.167.56.0/24** (Router 2's LAN for PC4/PC5). Without this route, packets destined for PC4/PC5 cannot be forwarded correctly.
Router 2:

- The routing table only shows:
  - **Connected networks:**
    - *192.167.56.0/24* (LAN connected to FastEthernet0/0 for PC4/PC5).
    - *12.0.0.0/8* (connected via Serial0/0/0 to Router 1).
  - **Static route:** *192.167.55.0/24* via *12.5.6.3* (Router 1).

Router 2 is also missing a route to **192.167.54.0/24** (Router 0's LAN for PC0/PC1). Therefore, packets from PC4/PC5 cannot reach PC0/PC1.

Steps to Fix the Issue
1. Add Missing Static Routes
Since static routes are being used, you need to add routes on both Router 0 and Router 2 to account for each other's LANs:
On Router 0:
config terminal
ip route 192.167.56.0 255.255.255.0 14.5.7.2
end
This tells Router 0 to forward packets for PC4/PC5 to Router 2's Serial interface at 14.5.7.2.
On Router 2:
config terminal
ip route 192.167.54.0 255.255.255.0 14.5.7.1
end
This tells Router 2 to forward packets for PC0/PC1 to Router 0's Serial interface at 14.5.7.1.

---

2. Verify Routes
After adding the static routes, verify the routing tables on both routers using show ip route. Ensure that:

- Router 0 has a route to *192.167.56.0/24* via *14.5.7.2*.
- Router 2 has a route to *192.167.54.0/24* via *14.5.7.1*.

---

3. Test Connectivity
Once the routes are in place:

1. Ping Router 2's FastEthernet0/0 interface (*192.167.56.3*) from Router 0.
2. Ping Router 0's FastEthernet0/0 interface (*192.167.54.1*) from Router 2.

3. Test end-to-end connectivity between:
    - PC0 and PC4.
    - PC1 and PC5.

Why It Didn't Work Previously
The missing static routes between Router 0 and Router 2 are the critical issue here. While the serial link between them is up, without explicit routing information, packets from one LAN cannot traverse the serial link to reach the other LAN.

Expected Result

Once the above steps are completed, PC0 and PC1 will be able to communicate with PC4 and PC5. Let me know how it goes, and I'll guide you further if needed!

You

(Now Thursday, Apr 10)
In Packet tracer I have a network setup with a router linking to 2 servers on ports Gig0/1 and Gig0/2 and ip 192.156.20.4 for server1 and ip 192.168.15.3 for server0 with a switch linked on port Gig0/0 and ip 192.168.78.1 that has 2 Laptops and a PC connected to it. How can I do this: configure a standard access control list (ACL) on the router to block traffic from the PC0 (IP: 192.168.78.3) to both the servers, while ensuring that all other devices can still communicate with the servers?

Copilot

To configure a **standard access control list (ACL)** on the router in Packet Tracer to block traffic from **PC0** (IP: 192.168.78.3) to both the servers, while allowing all other devices to communicate with the servers, follow these steps:

Steps to Configure Standard ACL
1. Access the Router
1. Open the **CLI (Command Line Interface)** for the router in Packet Tracer.
2. Enter privileged EXEC mode:
    enable
3. Enter global configuration mode:
    configure terminal

2. Create the ACL
1. Create a standard ACL that denies traffic from PC0 to both servers. Use the IP address of PC0 (*192.168.78.3*) to define the source traffic to block:
    access-list 10 deny 192.168.78.3
2. Add a rule to permit all other traffic (since ACLs end with an implicit deny all):
    access-list 10 permit any

3. Apply the ACL to the Interfaces
1. Apply the ACL inbound on the **GigabitEthernet0/1** interface (connected to Server 1 - IP: *192.156.20.4*):
    interface GigabitEthernet0/1
    ip access-group 10 in
    exit
2. Apply the ACL inbound on the **GigabitEthernet0/2** interface (connected to Server 0 - IP: *192.168.15.3*):
    interface GigabitEthernet0/2
    ip access-group 10 in
    exit

4. Verify the Configuration
1. Exit configuration mode:
    end
2. Save the configuration:
    write memory
3. Verify the ACLs applied on the interfaces:

show running-config

4. Test connectivity from **PC0** to the servers to confirm it is blocked. Test communication from other devices to ensure they can still reach the servers.

Explanation of the Configuration

- **Access-list 10 deny 192.168.78.3:** This denies traffic from PC0 to the servers. The standard ACL uses source IP addresses only to filter traffic.
- **Access-list 10 permit any:** This allows all other traffic.
- **ip access-group 10 in:** This applies the ACL to the interfaces where traffic to the servers enters the router.

By following this setup, **PC0's traffic to Server 1 and Server 0 will be blocked**, while other devices connected to the switch (like the laptops) will retain access.
Let me know if you need help testing or troubleshooting further!

You

I've set it up now (see attached screenshot [Expired]) but my ping command on PC0 (ip 192.168.78.3) still works

Copilot

From the configuration in the screenshot, it seems the **ACL (MyACL)** is correctly applied to the **GigabitEthernet0/1** and **GigabitEthernet0/2** interfaces, which are connected to the servers. However, the ping from **PC0** (IP: 192.168.78.3) still works because the ACL is applied **inbound** on the server interfaces, but the traffic from PC0 is likely passing through **GigabitEthernet0/0**, which does not have the ACL applied.

Here's how to fix this:

Steps to Resolve
1. Apply the ACL to GigabitEthernet0/0
Since PC0 is connected to the switch via **GigabitEthernet0/0**, the ACL needs to be applied **outbound** on this interface to block traffic from PC0 before it reaches the servers.
On the Router:
config terminal
interface GigabitEthernet0/0
ip access-group MyACL out
end
This ensures that traffic from PC0 is filtered as it exits the router towards the servers.

---

2. Verify the ACL Application
After applying the ACL, verify its placement using:
show running-config
Ensure that **MyACL** is applied to **GigabitEthernet0/0** in the **outbound direction**.

---

3. Test Connectivity

1. Ping the servers from **PC0** again. The ping should now fail.
2. Test connectivity from other devices (e.g., laptops) to ensure they can still communicate with the servers.

Why This Works

The ACL was initially applied **inbound** on the server interfaces (**GigabitEthernet0/1** and **GigabitEthernet0/2**), but this only filters traffic entering the router from those interfaces. Since PC0's traffic originates from **GigabitEthernet0/0**, it bypasses the ACL. Applying the ACL **outbound** on **GigabitEthernet0/0** ensures that PC0's traffic is blocked before it reaches the servers.

Let me know if this resolves the issue or if further troubleshooting is needed!

You

I've done that but I'm still getting the reply from the ping command [Uploaded Image Expired]

Copilot

If the ping from **PC0** (IP: 192.168.78.3) is still succeeding despite the ACL configuration, the issue might be with how traffic is being processed by the router, or with the placement of the ACL. Let's troubleshoot systematically and resolve this step-by-step.

Troubleshooting Steps

1. Confirm ACL Configuration
Make sure the ACL is correctly configured. Use the following command:

show access-list

You should see:

Standard IP access list 10
deny 192.168.78.3
permit any

If you don't see this, the ACL might not be applied or incorrectly configured. In that case, reapply the ACL to the interfaces.

---

2. Verify ACL Application
Check the router's interfaces to confirm that the ACL is applied correctly:

show running-config | include ip access-group

You should see:

GigabitEthernet0/0
ip access-group MyACL out

If the ACL isn't applied to the correct interface (GigabitEthernet0/0 outbound), reapply it:

config terminal
interface GigabitEthernet0/0
ip access-group MyACL out
end

---

3. Check the Direction of ACL Application
Ensure the ACL is applied in the correct direction:

- **Outbound** on **GigabitEthernet0/0** to filter traffic leaving the router towards the servers.

If the ACL is mistakenly applied **inbound**, traffic might bypass it entirely.

---

4. Test Traffic
Perform tests to isolate the issue:

- Ping the servers from **PC0** again and check if the ACL blocks the traffic.
- If the ping still works, test traffic to the servers using **other protocols** (like HTTP) to ensure the ACL is working.

---

5. Inspect Other Devices
It's possible that:

- Another device (like the switch) is interfering with packet filtering.
- The servers might have been configured to allow ICMP traffic explicitly, bypassing the ACL.

---

6. Final Checks
Use debugging commands to monitor traffic flow:

debug ip packet

Analyze whether packets from *192.168.78.3* are being denied by the ACL or bypassing it. Once done, disable debugging:

no debug all

Next Steps

1. Recheck ACL placement and direction.
2. Test with different protocols beyond ICMP (e.g., HTTP, SSH) to confirm the ACL behavior.
3. Let me know if you need guidance interpreting debug outputs or reapplying ACLs! Together, we'll nail it down!

You

How can I fix this, my network configuration is in the attached screenshot [Expired]?

Copilot

Based on the screenshots provided, here's a clear diagnosis and solution to fix your network configuration issue, ensuring that **PC0's traffic is blocked** from reaching the servers, while allowing all other devices to communicate freely:

Observations from the Screenshots

1. **Network Layout**:
   - **PC0 (192.168.78.3)** is connected via a switch to the router on **GigabitEthernet0/0**.
   - **Server0 (192.168.15.3)** and **Server1 (196.156.20.4)** are connected to the router via **GigabitEthernet0/1** and **GigabitEthernet0/2** respectively.
2. **ACL Configuration**:
   - The ACL *MyACL* has been applied to **GigabitEthernet0/0** in the **outbound direction**, but you were still able to ping the servers successfully from PC0.

Root Cause

While the ACL is correctly applied, **ICMP (ping) traffic might still be bypassing the ACL due to incorrect rules or incomplete filtering**. Standard ACLs only filter traffic based on **source IP addresses**, which might not be sufficient to block specific types of traffic.

Solution: Update the ACL Rules

Step 1: Modify the ACL

Reconfigure the ACL to ensure it explicitly blocks **all traffic** from PC0 to the servers:

```
config terminal
no access-list 10
access-list 10 deny 192.168.78.3
access-list 10 permit any
end
```

This rule blocks **all traffic originating from PC0**, while permitting traffic from other devices.

Step 2: Apply the Updated ACL

Ensure the ACL is applied **outbound** on **GigabitEthernet0/0**, which connects PC0 to the router. If already applied, recheck the configuration:

```
config terminal
interface GigabitEthernet0/0
ip access-group 10 out
end
```

Additionally, double-check that the ACL is **not applied inbound** on any interfaces linked to the servers (**GigabitEthernet0/1** and **GigabitEthernet0/2**), as this might result in unintended behavior.

Step 3: Verify the Configuration

1. Check the ACL rules:
   show access-list

Ensure the ACL reflects the updated rules.

2. Verify its placement on the router interfaces:
   show running-config | include ip access-group

Step 4: Test the Network

1. Ping **Server0** (*192.168.15.3*) and **Server1** (*196.156.20.4*) from PC0. The ping should now fail.
2. Ping the servers from Laptop0, Laptop1, and other devices—they should still have successful connectivity.
3. Test other protocols (HTTP, FTP, etc.) from PC0 to ensure all traffic is blocked.

Additional Considerations

If the ACL does not block **ICMP traffic**, you may need to use an **Extended ACL** instead of a Standard ACL. Extended ACLs allow finer control over traffic filtering, including specific protocols like ICMP. For example:

```
access-list 101 deny icmp host 192.168.78.3 any
access-list 101 permit ip any any
```

Apply this extended ACL similarly:

```
interface GigabitEthernet0/0
ip access-group 101 out
```

Expected Outcome
With this updated configuration:

- **PC0's traffic** (all protocols) to the servers will be blocked.
- All other devices will maintain unrestricted communication with the servers.

Let me know how the testing goes and if you need further assistance!

Conversation by Fifa Utuber with Copilot on Monday, Mar 24 2025